

Draft of Standard Operating Procedure (SOP) for evacuation of power from Solar Power Project through common line infrastructure

Standard Operating Procedure (SOP) for evacuation of power through common infrastructure from distinctive solar power project to be established under component A of PM KUSUM Scheme from pooling station and up to delivery/injection point at substation end: -

- a. Selection of RPG for injection power through common evacuation system at 33/11 KV substation based on the substation and capacity allotted to RPG by the Letter of Award (LoA) issued by MPUVNL.
- b. Injection of power from distinctive solar PV projects shall be connected to a pooling station before the power is injected into common evacuation system, and the power shall be injected mandatorily at 11 KV side of 33/11 KV substation subject to 80% capacity of the individual PTR.
- c. The levelized tariff determined by Hon'ble Commission has been arrived considering the cost of 5 KM 11 KV line from solar project to injection point and the in case of the availing benefit of power evacuation structure by the RPG in that case the impact of such arrangement shall be subject to review of the levelized tariff by Hon'ble MPERC.
- d. The distinctive RPGs shall install main & check metering arrangement having 0.2s accuracy class, tested at NABL accredited laboratory at the pooling station in accordance with the CEA guideline. Further main & check metering arrangement having 0.2s accuracy class, tested at NABL accredited laboratory shall be provided at point of injection/delivery point, i.e. 11 KV side of 33/11 KV Substation.
- e. For apportionment of power injected (export) by generator following methodology shall be applicable: -

For example:

If energy generated by RPG 1 is X units and energy generated by RPG 2 is Y units. Energy recorded at injection point/ delivery point at 33/11 KV substation end is Z units and transmission losses are L units. Then,

$$L = (X+Y)-Z$$

$$\text{Energy generation unit to be billed for RPG 1} = X - \frac{X \cdot L}{(X+Y)}$$

$$\text{Energy generation unit to be billed for RPG 2} = Y - \frac{Y \cdot L}{(X+Y)}$$

- f. For apportionment of power drawn from the grid (import) during non-generation period following methodology shall be applicable: -

For example:

If energy recorded in meter of RPG 1 at pooling station is X units and energy recorded in meter of RPG 2 at pooling station is Y units. Energy supplied by DISCOM from substation is Z unit and transmission losses are L Unit. Then,

$$L = Z - (X+Y)$$

Energy consumed unit to be billed for RPG 1 = $X + X \cdot L / (X + Y)$

Energy consumed unit to be billed for RPG 2 = $Y + Y \cdot L / (X + Y)$

- g. The distinctive generators shall submit a common undertaking jointly signed and irrevocably authorizing one of the constituent RPG, same shall be responsible for apportionment & other common issues, indemnifying MPPMCL and the DISCOM for any consequential legal, financial, commercial, safety, as well as maintenance issues pertaining to common evacuation system.
- h. Maintenance/repairing/replacement of the common evacuation system and metering arrangement shall be the responsibility of both the RPGs.
- i. The reading of the metering arrangement installed at generator's end shall be recorded jointly by both the RPGs. After Joint Meter Reading, JMR statement shall be prepared and signed by both the RPGs and person authorized by respective SE/GM of circle.
- j. In case any dispute arising between RPGs, the decision of the concerned Superintending Engineer/ General Manager of O&M Circle under whose jurisdiction of substation and RPGs projects are installed shall be final and binding.
- k. The invoice submitted by individual RPG shall be in accordance with apportionment arrangement agreed between RPGs and as approved by the jurisdiction in correspondence with the JMR recorded by check and main metering arrangement provided at injection point and pooling station.
- l. The conductor size should be in accordance with the maximum combined loading capacity generated from solar power project. To guide broad contours of conductor size, viz length of line from substation to solar power plant are as under: -

Sr. No.	Length of 11 kV Line (in Km)	Solar power injection capacity (in MW)	ACSR conductor size	Support
1.	up to 4 km	up to 4 MW	75 sq mm (Raccoon)	PCC pole 280 kg. 9.1 m
		Above 4 MW	100 sq mm (Dog)	H-Beam
2.	Above 4 km	up to 4 MW	75 sq mm (Raccoon)	H-Beam
		Above 4 MW	100 sq mm (Dog)	H-Beam